TUTORIAL 10

Similarity and Determinants

MOTIVATION AND GUIDING IDEA

A change of basis preserves the linear map but can greatly simplify its matrix. The determinant tests invertibility without computing an inverse, detects dependence among rows, and measures changes in area or volume. Before a long calculation, look for zeros, linear dependence, or block structure.

NOTATION AND CONVENTIONS

- $M_n(\mathbb{F})$ is the space of square matrices of size n over $\mathbb{F}$; I_n is the identity matrix, and A^T is the transpose.
- $A \sim B$ means that $B = P^{-1}AP$ for some invertible P. The columns of P are the new basis vectors expressed in the old basis.
- $\det A$ is the determinant. The alternative notation $|A|$ sometimes means the same scalar; it is not an absolute value of a matrix.
- M_{ij} is obtained by deleting row i and column j; $C_{ij} = (-1)^{i+j} \det(M_{ij})$ is the corresponding cofactor.
- $R_i \leftarrow R_i + cR_j$ means replacing row i by itself plus c times row j. The determinant preservation rule requires $i \neq j$.

DEFINITION 10.1

For every $n \geq 1$, the determinant is the unique function $\det : M_n(\mathbb{F}) \to \mathbb{F}$ that is linear in each row separately with the other rows fixed, vanishes when two rows are equal, and satisfies $\det I_n = 1$. Row operation rules and cofactor expansions follow. Expansion along row i gives $\det A = \sum_{j=1}^{n} a_{ij}C_{ij}$; the result is independent of the selected row. With the convention $\det(\varnothing) = 1$ for the empty matrix of size 0, the formula also holds for $n = 1$.

NOTE

For square matrices over any field, $\det(AB) = \det A \det B$, $\det(A^T) = \det A$, and $\det(cA) = c^n \det A$. Also, A is invertible exactly when $\det A \neq 0$; then $\det(A^{-1}) = (\det A)^{-1}$. The determinant is not generally additive: $\det(A + B) \neq \det A + \det B$ in general.

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NOTE

Distinguish two transformations. Similarity, $B = P^{-1}AP$, describes an operator after the same basis change in domain and codomain; it preserves the determinant. Congruence, $B = P^TAP$, describes a basis change for a bilinear form; here $\det B = (\det P)^2 \det A$. The formulas cannot be interchanged.

NOTE

The geometric interpretation requires a real matrix. In $\mathbb{R}^2$, $|\det A|$ is the area scale factor; in $\mathbb{R}^3$, it is the volume scale factor. If $\det A = 0$, the image has smaller dimension. If $\det A \neq 0$, the sign indicates whether orientation is preserved or reversed.

MOTIVATION AND GUIDING IDEA

By the end of this tutorial, you will be able to choose an efficient determinant calculation, track the effect of row operations, identify conditions for invertibility, and connect algebraic structure with changes of basis and area.

10.1 Similar Matrices

RECALL AND TOOLS

Matrices $A, B \in \mathbb{F}^{n\times n}$ are called similar if there is an invertible matrix $P \in \mathbb{F}^{n\times n}$ such that

$$A = P^{-1}BP$$

If A, B are similar, then

1. $$\det(A) = \det(B)$$
2. $$\operatorname{trace}(A) = \operatorname{trace}(B)$$
3. $$\operatorname{rank}(A) = \operatorname{rank}(B)$$

EXERCISE 10.1

Let $A, B \in \mathbb{F}^{n\times n}$. Please prove or disprove:

1. If A, B are similar, then A^k, B^k are similar for every integer $k \geq 1$.
2. If A^k, B^k are similar for some integer $k \geq 1$, then A, B are similar.

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EXERCISE 10.1 CONTINUED

WORKED SOLUTION

1. If $B = P^{-1}AP$ with P invertible, cancellation using $PP^{-1} = I$ gives

$$B^k = (P^{-1}AP)^k = P^{-1}A^kP.$$

   Thus the powers are similar via the same matrix P.

2. The statement is false. Over any field, take

$$A = 0_{2\times 2}, \qquad B = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

   We have $A^2 = B^2 = 0$, but A, B are not similar: $P^{-1}AP = 0$ for every invertible P, whereas $B \neq 0$. Over $\mathbb{R}$ or $\mathbb{C}$, the original example $A = -I_2$, $B = I_2$ also works: their squares are equal, and a scalar matrix is similar only to itself. This example gives no counterexample in characteristic 2, where $-I_2 = I_2$.

Common pitfall: The equality $A^k = (Q^{-1}BQ)^k$ does not imply $A = Q^{-1}BQ$. Taking a power is not an injective operation on matrices, as these examples show.

10.2 The Determinant

The determinant is defined for square matrices. In the real plane, its absolute value is the area scale factor; its sign indicates whether orientation is preserved or reversed.

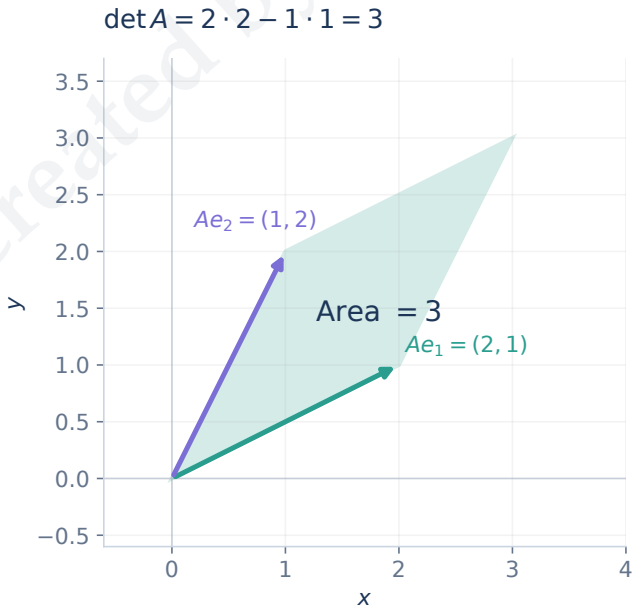


Figure 10.1. A linear map sends the unit square to a parallelogram. The parallelogram's area is the absolute value of the determinant.

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DEFINITION 10.2

Let $n \in \mathbb{N}$. A function

$$\det : M_n(\mathbb{R}) \longrightarrow \mathbb{R}$$

is called the **determinant** if, for every $A \in M_n(\mathbb{R})$, it has the following properties:

1. **Interchanging rows (change of sign):** For $1 \leq i < j \leq n$, if $E^{(1)}_{i,j} \in M_n(\mathbb{R})$ is the elementary matrix interchanging rows i and j, then
$$\det\left(E^{(1)}_{i,j} A\right) = -\det(A).$$
2. **Multiplying a row by a scalar (homogeneity):** For $1 \leq i \leq n$ and $\lambda \in \mathbb{R} \setminus \{0\}$, if $E^{(2)}_{i,\lambda} \in M_n(\mathbb{R})$ is the elementary matrix multiplying row i by λ, then
$$\det\left(E^{(2)}_{i,\lambda} A\right) = \lambda \det(A).$$
3. **Adding a multiple of another row (invariance):** For $1 \leq i \neq j \leq n$ and $\lambda \in \mathbb{R}$, if $E^{(3)}_{i,j,\lambda} \in M_n(\mathbb{R})$ is the elementary matrix performing $R_i \leftarrow R_i + \lambda R_j$, then
$$\det\left(E^{(3)}_{i,j,\lambda} A\right) = \det(A).$$
4. **Normalization:**
$$\det(I_n) = 1.$$

RECALL AND TOOLS

Two useful basic properties:

- If $A = (a_{ij}) \in M_n(\mathbb{R})$ is upper triangular, meaning that $a_{ij} = 0$ for every $i > j$, then
$$\det(A) = \prod_{k=1}^{n} a_{kk}.$$
- If rows i and j are equal for some $i \neq j$, or if A has a zero row, then $\det(A) = 0$. Similarly, two equal columns or a zero column imply $\det(A) = 0$.

EXERCISE 10.2

Please compute the determinants using row operations, stating the effect of each operation on the determinant.

1. For $t \in \mathbb{R}$,
$$A(t) = \begin{pmatrix} 1 & 1 & 0 & -1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 \\ 2 & 1 & t & 7 \end{pmatrix}.$$

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EXERCISE 10.2 CONTINUED

2.
$$B = \begin{pmatrix} 2 & 1 & 0 & 3 & -1 \\ 0 & 1 & 2 & -1 & 1 \\ 2 & -1 & -4 & 5 & -3 \\ 1 & 0 & 1 & 0 & 2 \\ 0 & 2 & 0 & 1 & 1 \end{pmatrix}.$$

WORKED SOLUTION

1. Add multiples of other rows to the fourth row; these operations preserve the determinant:
$$A(t) \xrightarrow{R_4 \leftarrow R_4 - 2R_1} \begin{pmatrix} 1 & 1 & 0 & -1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & -1 & t & 9 \end{pmatrix}$$
$$\xrightarrow{R_4 \leftarrow R_4 + R_2} \begin{pmatrix} 1 & 1 & 0 & -1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & t+2 & 10 \end{pmatrix}$$
$$\xrightarrow{R_4 \leftarrow R_4 - (t+2)R_3} \begin{pmatrix} 1 & 1 & 0 & -1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 6-2t \end{pmatrix}.$$
Multiplying the diagonal entries gives $\det A(t) = 1 \cdot 1 \cdot 1 \cdot (6 - 2t) = 6 - 2t$.

2. The rows of B satisfy $r_3 = r_1 - 2r_2$. Thus the two operations $R_3 \leftarrow R_3 - R_1$ and $R_3 \leftarrow R_3 + 2R_2$ produce a zero row without changing the determinant. A matrix with a zero row has determinant zero, so $\det B = 0$.

RECALL AND TOOLS

Basic properties (from the lectures):

- **Multiplicativity:** For every $A, B \in M_n(\mathbb{R})$,
$$\det(AB) = \det(A)\det(B).$$
- **Failure of additivity (in general):** In general, the following equality does not hold:
$$\det(A + B) = \det(A) + \det(B).$$
- **Invariance under transposition:** For every $A \in M_n(\mathbb{R})$,
$$\det(A^T) = \det(A).$$
- **Scaling the entire matrix:** For every $c \in \mathbb{R}$ and $A \in M_n(\mathbb{R})$,
$$\det(cA) = c^n \det(A),$$

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where n is the size of the matrix.

Fundamental theorem:

PROPOSITION 10.1

For every $A \in M_n(\mathbb{R})$, A is invertible if and only if $\det(A) \neq 0$.

EXERCISE 10.3

Let $A, B \in M_n(\mathbb{R})$, with A singular and B invertible. Please prove or disprove that the matrix in each part is singular:

1. $A^T B$.
2. $B^{-1}AB$.
3. $A + A^2 + A^3 + A^4$.
4. $(A + I)^2 - (A - I)^2$.
5. $(A + B)^2 + (A - B)^2$.

WORKED SOLUTION

Use $\det A = 0$, $\det B \neq 0$, and multiplicativity of the determinant.

1. $\det(A^T B) = \det(A^T)\det B = \det A \det B = 0$.
2. $\det(B^{-1}AB) = \det(B^{-1})\det A \det B = 0$.
3. Factor out A on the left:

$$A + A^2 + A^3 + A^4 = A(I + A + A^2 + A^3),$$

Therefore the determinant is zero.

4. Since $AI = IA = A$, we obtain

$$(A + I)^2 - (A - I)^2 = (A^2 + 2A + I) - (A^2 - 2A + I) = 4A.$$

Consequently, $\det(4A) = 4^n \det A = 0$.

5. Expand while preserving the order of multiplication:

$$(A + B)^2 + (A - B)^2 = (A^2 + AB + BA + B^2) + (A^2 - AB - BA + B^2) = 2(A^2 + B^2).$$

The determinant is therefore $2^n \det(A^2 + B^2)$, but it need not vanish. A counterexample is $A = 0$, $B = I_n$: the resulting matrix is $2I_n$, which is invertible and has determinant $2^n \neq 0$.

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10.2.1 The Determinant Formula

RECALL AND TOOLS

Let $A = (a_{ij}) \in M_n(\mathbb{R})$ be a square matrix of size n. For $1 \leq i, j \leq n$, the **minor matrix** corresponding to entry (i, j), denoted M_{ij}, is the square matrix of size $n-1$ obtained by deleting row i and column j from A.

EXAMPLE 10.1

Let $A = \begin{pmatrix} 9 & 8 & 7 \\ 6 & 5 & 4 \\ 3 & 2 & 1 \end{pmatrix}$. Then $M_{11} = \begin{pmatrix} 5 & 4 \\ 2 & 1 \end{pmatrix}$ and $M_{23} = \begin{pmatrix} 9 & 8 \\ 3 & 2 \end{pmatrix}$.

DEFINITION 10.3

Let $A = (a_{ij}) \in M_n(\mathbb{R})$ be square of size n. Its **determinant**, denoted $\det(A)$, can be computed recursively in n:

1. If $n = 1$, then $A = (a_{11})$ and we define
$$\det(A) = a_{11}.$$
2. If $n = 2$, then
$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}, \qquad \det(A) = a_{11}a_{22} - a_{12}a_{21}.$$
3. If $n \geq 3$, then for every $1 \leq i \leq n$, expansion along row i gives
$$\det(A) = \sum_{j=1}^{n} a_{ij} \, (-1)^{i+j} \, \det(M_{ij}),$$
where M_{ij} is obtained by deleting row i and column j.
Equivalently, for every $1 \leq j \leq n$, expansion along column j gives
$$\det(A) = \sum_{i=1}^{n} a_{ij} \, (-1)^{i+j} \, \det(M_{ij}).$$

EXERCISE 10.4

Please compute the determinants again, this time using cofactor expansion and its consequences.

1. For $t \in \mathbb{R}$,
$$A(t) = \begin{pmatrix} 1 & 1 & 0 & -1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 \\ 2 & 1 & t & 7 \end{pmatrix}.$$

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EXERCISE 10.4 CONTINUED

2.
$$B = \begin{pmatrix} 2 & 1 & 0 & 3 & -1 \\ 0 & 1 & 2 & -1 & 1 \\ 2 & -1 & -4 & 5 & -3 \\ 1 & 0 & 1 & 0 & 2 \\ 0 & 2 & 0 & 1 & 1 \end{pmatrix}.$$

WORKED SOLUTION

1. Expand along the first column. Its two nonzero entries are in positions $(1,1)$ and $(4,1)$, so
$$\det A(t) = \det M_{11} - 2\det M_{41}.$$
The minor matrices are
$$M_{11} = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 2 \\ 1 & t & 7 \end{pmatrix}, \qquad M_{41} = \begin{pmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}.$$
Expand the first along its first column and the second along its first row:
$$\det M_{11} = (1\cdot 7 - 2t) + (2\cdot 2 - 1\cdot 1) = 10 - 2t,$$
$$\det M_{41} = 1(2\cdot 2 - 1\cdot 1) + (-1)(1\cdot 1 - 2\cdot 0) = 3 - 1 = 2.$$
Thus $\det A(t) = 10 - 2t - 2\cdot 2 = 6 - 2t$, as obtained by the previous method.

2. Regard the determinant as a function of the rows. Since $r_3 = r_1 - 2r_2$, linearity in the third row gives
$$\begin{aligned}\det(r_1, r_2, r_1 - 2r_2, r_4, r_5) = {} & \det(r_1, r_2, r_1, r_4, r_5) \\ & - 2\det(r_1, r_2, r_2, r_4, r_5) = 0.\end{aligned}$$
Each term has two equal rows. Linearity in a row and vanishing for equal rows follow from the determinant expansion, so again $\det B = 0$.

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EXERCISE 10.5

Using the determinant, please find all $k \in \mathbb{R}$ for which
$$A(k) = \begin{pmatrix} 2 & k & 4 \\ 3 & k+1 & 6 \\ -1 & -1 & k+1 \end{pmatrix}$$
is singular.

WORKED SOLUTION

A square matrix is singular exactly when its determinant is zero. Expand along the first row:
$$\begin{aligned}\det A(k) = {} & 2\big((k+1)^2 + 6\big) - k\big(3(k+1) + 6\big) \\ & + 4\big(-3 + (k+1)\big)\end{aligned}$$

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EXERCISE 10.5 CONTINUED

$$= 2k^2 + 4k + 14 - 3k^2 - 9k + 4k - 8$$
$$= -k^2 - k + 6 = -(k+3)(k-2).$$

Hence the required values are $k = -3, 2$. For every other value, the determinant is nonzero, so the matrix is invertible.

EXERCISE 10.6

1. Let $n \geq 2$ and let $x_1, \ldots, x_n$ be real (or complex) numbers. Define the Vandermonde matrix

$$V(x_1, \ldots, x_n) = \left(x_i^{j-1}\right)_{i,j=1}^{n} = \begin{pmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^{n-1} \\ 1 & x_2 & x_2^2 & \cdots & x_2^{n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^{n-1} \end{pmatrix}.$$

Please prove that

$$\det\big(V(x_1, \ldots, x_n)\big) = \prod_{1 \leq i < j \leq n} (x_j - x_i).$$

WORKED SOLUTION

1. Write

$$\Delta_n(x_1, \ldots, x_n) = \det\big(V(x_1, \ldots, x_n)\big).$$

We prove by induction on n that

$$\Delta_n(x_1, \ldots, x_n) = \prod_{1 \leq i < j \leq n} (x_j - x_i).$$

Base case, $n = 2$:

$$\Delta_2(x_1, x_2) = \det \begin{pmatrix} 1 & x_1 \\ 1 & x_2 \end{pmatrix} = x_2 - x_1,$$

as required.

Inductive step: assume the statement holds for $n-1$, and prove it for n. Regard Δ_n as a polynomial in x_n, with $x_1, \ldots, x_{n-1}$ fixed. Expanding the determinant along row n gives

$$\Delta_n(x_1, \ldots, x_n) = \sum_{j=1}^{n} x_n^{j-1} \, (-1)^{n+j} \, \det(M_{nj}),$$

Thus Δ_n is a polynomial in x_n of degree at most $n-1$. The only term containing x_n^{n-1} comes from column n, so its coefficient is

$$(-1)^{n+n} \det(M_{nn}) = \det\big(V(x_1, \ldots, x_{n-1})\big) = \Delta_{n-1}(x_1, \ldots, x_{n-1}).$$

If two of $x_1, \ldots, x_{n-1}$ are equal, the matrix has two equal rows, and both sides of the desired formula are zero. Hence we may assume these numbers are pairwise distinct. For every $k \in \{1, \ldots, n-1\}$, substituting $x_n = x_k$ makes rows k and n of V equal, so

$$\Delta_n(x_1, \ldots, x_{n-1}, x_k) = 0.$$

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EXERCISE 10.6 CONTINUED

Therefore $(x_n - x_k)$ divides Δ_n for every $k = 1, \ldots, n-1$ (as a polynomial in x_n). Since these roots are distinct and the degree is at most $n-1$, there is a polynomial P such that

$$\Delta_n(x_1, \ldots, x_n) = \Big(\prod_{k=1}^{n-1}(x_n - x_k)\Big) P(x_1, \ldots, x_{n-1}),$$

where P is independent of x_n.

Compare the coefficients of x_n^{n-1} on both sides. On the left the coefficient is $\Delta_{n-1}(x_1, \ldots, x_{n-1})$; on the right the coefficient of x_n^{n-1} in $\prod_{k=1}^{n-1}(x_n - x_k)$ is 1. Hence

$$P(x_1, \ldots, x_{n-1}) = \Delta_{n-1}(x_1, \ldots, x_{n-1}).$$

It follows that

$$\Delta_n(x_1, \ldots, x_n) = \Delta_{n-1}(x_1, \ldots, x_{n-1}) \prod_{k=1}^{n-1}(x_n - x_k).$$

Finally, by the induction hypothesis,

$$\Delta_{n-1}(x_1, \ldots, x_{n-1}) = \prod_{1 \le i < j \le n-1} (x_j - x_i),$$

and therefore

$$\Delta_n(x_1, \ldots, x_n) = \Big(\prod_{1 \le i < j \le n-1} (x_j - x_i)\Big)\Big(\prod_{k=1}^{n-1}(x_n - x_k)\Big) = \prod_{1 \le i < j \le n} (x_j - x_i).$$

This proves the formula.

EXERCISE 10.7

1. Let $n \ge 2$ and let $w \in \mathbb{C}$ be a primitive nth root of unity: $w^n = 1$ and $w^k \ne 1$ for every $1 \le k \le n-1$. Define the matrix

$$F = \left(w^{(i-1)(j-1)}\right)_{i,j=1}^{n} \in M_n(\mathbb{C}).$$

Please compute $\det(F)$ explicitly.

2. Let $n \ge 3$, let $1 \le p \le n-1$, and please write $q = n - p$. For $\alpha, \beta \in \mathbb{R}$, define $A(\alpha, \beta) \in M_n(\mathbb{R})$ by

$$a_{ii} = 1 \quad (1 \le i \le n),$$

and, for every $i \ne j$,

$$a_{ij} = \begin{cases} \alpha, & 1 \le i, j \le p \text{ or } p+1 \le i, j \le n, \\ \beta, & \text{otherwise.} \end{cases}$$

Let $T : \mathbb{R}^n \to \mathbb{R}^n$ be the linear map $T(v) = A(\alpha, \beta)v$. Please determine for which α, β the map T is an isomorphism, and please compute $\det(A(\alpha, \beta))$.

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EXERCISE 10.7 CONTINUED

WORKED SOLUTION

First display the structure of the matrices:

$$F = \left(w^{(i-1)(j-1)}\right)_{i,j=1}^{n} = \begin{pmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & w & w^2 & \cdots & w^{n-1} \\ 1 & w^2 & w^4 & \cdots & w^{2(n-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & w^{n-1} & w^{2(n-1)} & \cdots & w^{(n-1)(n-1)} \end{pmatrix}.$$

$$A(\alpha,\beta) = \left(\begin{array}{c|c} A_{11} & A_{12} \\ \hline A_{21} & A_{22} \end{array}\right),.$$

The block sizes are

$$A_{11} \in M_p(\mathbb{R}), \qquad A_{22} \in M_q(\mathbb{R}),$$
$$A_{12} \in M_{p\times q}(\mathbb{R}), \qquad A_{21} \in M_{q\times p}(\mathbb{R}).$$

$$A_{11} = \begin{pmatrix} 1 & \alpha & \alpha & \cdots & \alpha \\ \alpha & 1 & \alpha & \cdots & \alpha \\ \alpha & \alpha & 1 & \cdots & \alpha \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \alpha & \alpha & \alpha & \cdots & 1 \end{pmatrix}_{p\times p}, \qquad A_{22} = \begin{pmatrix} 1 & \alpha & \alpha & \cdots & \alpha \\ \alpha & 1 & \alpha & \cdots & \alpha \\ \alpha & \alpha & 1 & \cdots & \alpha \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \alpha & \alpha & \alpha & \cdots & 1 \end{pmatrix}_{q\times q}.$$

$$A_{12} = \begin{pmatrix} \beta & \beta & \cdots & \beta \\ \beta & \beta & \cdots & \beta \\ \vdots & \vdots & \ddots & \vdots \\ \beta & \beta & \cdots & \beta \end{pmatrix}_{p\times q}, \qquad A_{21} = \begin{pmatrix} \beta & \beta & \cdots & \beta \\ \beta & \beta & \cdots & \beta \\ \vdots & \vdots & \ddots & \vdots \\ \beta & \beta & \cdots & \beta \end{pmatrix}_{q\times p}.$$

$$P^{-1}A(\alpha,\beta)P = (1-\alpha)I_{n-2} \oplus B = \left(\begin{array}{c|c} (1-\alpha)I_{n-2} & 0 \\ \hline 0 & B \end{array}\right),$$

where the lower-right block has size 2:

$$B = \begin{pmatrix} 1+(p-1)\alpha & q\beta \\ p\beta & 1+(q-1)\alpha \end{pmatrix}.$$

1. **Fact used below (Vandermonde determinant).** For $x_1,\dots,x_n \in \mathbb{C}$, define

$$V(x_1,\dots,x_n) = \left(x_i^{j-1}\right)_{i,j=1}^{n}.$$

Then

$$\det\big(V(x_1,\dots,x_n)\big) = \prod_{1\le i<j\le n} (x_j - x_i).$$

Set

$$x_i = w^{i-1} \qquad (i = 1,\dots,n).$$

For every i,j,

$$F_{ij} = w^{(i-1)(j-1)} = (w^{i-1})^{j-1} = x_i^{j-1}.$$

Therefore

$$F = V(x_1,\dots,x_n).$$

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EXERCISE 10.7 CONTINUED

By the preceding exercise on the Vandermonde determinant,

$$\det(F) = \prod_{1\le i<j\le n} (x_j - x_i) = \prod_{0\le i<j\le n-1} \left(w^j - w^i\right).$$

For every $0 \le i < j \le n-1$,

$$w^j - w^i = w^i\left(w^{j-i} - 1\right),$$

so

$$\det(F) = \left(\prod_{0\le i<j\le n-1} w^i\right)\left(\prod_{0\le i<j\le n-1} (w^{j-i} - 1)\right).$$

Write

$$S = \sum_{0\le i<j\le n-1} i.$$

Then

$$\prod_{0\le i<j\le n-1} w^i = w^S.$$

We now compute S: For each $i \in \{0, \dots, n-2\}$, there are exactly $(n-1-i)$ values of j with $i < j \le n-1$, so

$$S = \sum_{i=0}^{n-2} i(n-1-i) = \frac{n(n-1)(n-2)}{6}.$$

Consequently,

$$\prod_{0\le i<j\le n-1} w^i = w^{\frac{n(n-1)(n-2)}{6}}.$$

Also group the factors by $m = j - i \in \{1, \dots, n-1\}$. For each m, there are exactly $(n-m)$ pairs (i, j) with $j - i = m$, so

$$\prod_{0\le i<j\le n-1} (w^{j-i} - 1) = \prod_{m=1}^{n-1} (w^m - 1)^{n-m}.$$

Therefore

$$\boxed{\det(F) = w^{\frac{n(n-1)(n-2)}{6}} \prod_{m=1}^{n-1} (w^m - 1)^{n-m}}.$$

Because w is a primitive nth root of unity, $w^m \ne 1$ for $1 \le m \le n-1$, so $w^m - 1 \ne 0$. Every factor in the product is therefore nonzero, and $\det(F) \ne 0$.

2. **Theorems used below (from class).**

(a) Isomorphisms and representing matrices. Let V be a finite-dimensional real vector space, let $T : V \to V$ be linear, and let $[T]_{\mathcal{B}}$ represent T in a basis $\mathcal{B}$. Then

$$T \text{ is an isomorphism } \iff [T]_{\mathcal{B}} \text{ is invertible.}$$

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EXERCISE 10.7 CONTINUED

(b) Invertibility and the determinant. For every $A \in M_n(\mathbb{R})$,

$$A \text{ is invertible } \iff \det(A) \neq 0.$$

(This is the standard determinant criterion for invertibility.)

(c) Invariance of the determinant under change of basis. If P is invertible, then

$$\det(P^{-1}AP) = \det(A),$$

because

$$\det(P^{-1}AP) = \det(P^{-1})\det(A)\det(P) = \det(A).$$

We can now solve the problem.

Write $A = A(\alpha, \beta)$. Define

$$U = \Big\{x \in \mathbb{R}^n : x_{p+1} = \cdots = x_n = 0, \quad \sum_{i=1}^{p} x_i = 0\Big\},$$

$$W = \Big\{x \in \mathbb{R}^n : x_1 = \cdots = x_p = 0, \quad \sum_{i=p+1}^{n} x_i = 0\Big\}.$$

Clearly U, W are subspaces, since they are defined by homogeneous linear equations in the coordinates.

We first prove that $Ax = (1-\alpha)x$ for every $x \in U$. Let $x \in U$. Then $x_{p+1} = \cdots = x_n = 0$ and $\sum_{j=1}^{p} x_j = 0$.

If $1 \leq i \leq p$, the structure of row i of A (diagonal entry 1, entries α elsewhere in its block, and entries β between blocks) gives

$$(Ax)_i = 1 \cdot x_i + \sum_{\substack{1 \leq j \leq p \\ j \neq i}} \alpha x_j + \sum_{j=p+1}^{n} \beta x_j.$$

The last sum vanishes because $x_{p+1} = \cdots = x_n = 0$, so

$$(Ax)_i = x_i + \alpha \sum_{\substack{1 \leq j \leq p \\ j \neq i}} x_j.$$

Now

$$\sum_{\substack{1 \leq j \leq p \\ j \neq i}} x_j = \sum_{j=1}^{p} x_j - x_i = 0 - x_i = -x_i,$$

and hence

$$(Ax)_i = x_i + \alpha(-x_i) = (1-\alpha)x_i.$$

If $p+1 \leq i \leq n$, the coefficients of coordinates $1, \ldots, p$ in this row are all β (the second block of x is zero). Thus

$$(Ax)_i = \sum_{j=1}^{p} \beta x_j = \beta \sum_{j=1}^{p} x_j = 0 = (1-\alpha)x_i.$$

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EXERCISE 10.7 CONTINUED

Therefore $Ax = (1-\alpha)x \in U$. In particular, U is A-invariant, and the restriction $A|_U$ is multiplication by $(1-\alpha)$. The same argument gives $Ay = (1-\alpha)y$ for every $y \in W$. Thus W is also A-invariant and $A|_W = (1-\alpha)\mathrm{Id}_W$.

Define

$$u = (\underbrace{1,\dots,1}_{p},\underbrace{0,\dots,0}_{q}), \qquad v = (\underbrace{0,\dots,0}_{p},\underbrace{1,\dots,1}_{q}), \qquad S = \mathrm{span}\{u,v\}.$$

We next compute Au and Av.

For every $1 \le i \le p$,

$$(Au)_i = 1\cdot 1 + \sum_{\substack{1\le j\le p\\ j\ne i}} \alpha\cdot 1 + \sum_{j=p+1}^{n} \beta\cdot 0 = 1 + (p-1)\alpha.$$

and, for every $p+1 \le i \le n$,

$$(Au)_i = \sum_{j=1}^{p} \beta\cdot 1 + \Big(1\cdot 0 + \sum_{\substack{p+1\le j\le n\\ j\ne i}} \alpha\cdot 0\Big) = p\beta.$$

Therefore

$$Au = \big(1+(p-1)\alpha\big)u + p\beta\, v.$$

Similarly,

$$Av = q\beta\, u + \big(1+(q-1)\alpha\big)v.$$

Thus S is A-invariant, and its representing matrix in the basis (u,v) is

$$B = \begin{pmatrix} 1+(p-1)\alpha & q\beta \\ p\beta & 1+(q-1)\alpha \end{pmatrix}.$$

We claim that

$$\mathbb{R}^n = (U\oplus W)\oplus S.$$

First, $U\cap W = \{0\}$, since a vector in both subspaces must have every coordinate zero. Also, if $au+bv \in U\oplus W$, the sum of its first-block coordinates is $ap=0$, so $a=0$; the sum of its second-block coordinates is $bq=0$, so $b=0$. Hence $S\cap(U\oplus W)=\{0\}$.

Next, every $x\in\mathbb{R}^n$ admits a decomposition $x = x_{UW} + x_S$ with $x_{UW}\in U\oplus W$ and $x_S \in S$. Write

$$s_1 = \sum_{i=1}^{p} x_i, \qquad s_2 = \sum_{i=p+1}^{n} x_i,$$

and define

$$x_S = \frac{s_1}{p}u + \frac{s_2}{q}v \in S, \qquad x_{UW} = x - x_S.$$

The sum of the first-block coordinates of x_{UW} is $s_1 - s_1 = 0$, and the second-block sum is $s_2 - s_2 = 0$, so $x_{UW}\in U\oplus W$. This proves the direct-sum decomposition.

Choose a basis $\mathcal{B}$ by concatenating bases of U and W and the basis (u,v) of S. In this

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EXERCISE 10.7 CONTINUED

basis, the matrix of A has the block diagonal form

$$(1-\alpha)I_{n-2} \oplus B.$$

Let P be the change-of-coordinates matrix from $\mathcal{B}$ to the standard basis. Then

$$P^{-1}AP = (1-\alpha)I_{n-2} \oplus B.$$

Combining facts (b) and (c),

$$\det(A) = \det(P^{-1}AP) = \det\big((1-\alpha)I_{n-2} \oplus B\big).$$

The determinant of a block diagonal matrix is the product of the determinants of its blocks (this follows by cofactor expansion, or by a suitable factorization and multiplicativity $\det(AB) = \det(A)\det(B)$). Therefore

$$\det(A) = \det\big((1-\alpha)I_{n-2}\big) \cdot \det(B) = (1-\alpha)^{n-2}\det(B).$$

Compute $\det(B)$:

$$\det(B) = \big(1+(p-1)\alpha\big)\big(1+(q-1)\alpha\big) - pq\,\beta^2.$$

and hence

$$\boxed{\det(A) = (1-\alpha)^{n-2}\Big(\big(1+(p-1)\alpha\big)\big(1+(q-1)\alpha\big) - pq\,\beta^2\Big)}.$$

By fact (a), T is an isomorphism if and only if its representing matrix A in the standard basis is invertible. By fact (b), this is equivalent to $\det(A) \neq 0$. Thus

$$T \text{ is an isomorphism} \iff \det(A) \neq 0 \iff (1-\alpha)^{n-2} \neq 0 \text{ and } \det(B) \neq 0.$$

Since $\alpha, \beta \in \mathbb{R}$ and $n \geq 3$, $(1-\alpha)^{n-2} \neq 0$ holds if and only if $\alpha \neq 1$. The exact condition is therefore

$$\boxed{\alpha \neq 1} \quad \text{and} \quad \boxed{\big(1+(p-1)\alpha\big)\big(1+(q-1)\alpha\big) \neq pq\,\beta^2}.$$

EXERCISE 10.8

For $s \in \mathbb{R}$, define $T_s : \mathbb{R}^2 \to \mathbb{R}^2$ by

$$T_s(v) = A_s v, \qquad A_s = \begin{pmatrix} 2 & 1 \\ s & 2 \end{pmatrix}.$$

The triangle Δ has vertices $(0,0)$, $(1,0)$, $(0,1)$. Please find the vertices and area of its image, please determine when T_s is invertible, and please describe the image when $s = 4$.

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EXERCISE 10.8 CONTINUED

WORKED SOLUTION

The images of the vertices are $(0,0)$, $(2,s)$, $(1,2)$. The original triangle has area $1/2$, and the area scale factor is $|\det A_s| = |2\cdot 2 - 1\cdot s| = |4-s|$. Hence the image has area $|4-s|/2$. Also, T_s is invertible exactly when $4-s\neq 0$, namely when $s\neq 4$. When $s=4$, the columns satisfy $(2,4)^T = 2(1,2)^T$. Every image is therefore a multiple of $(1,2)^T$, and the triangle collapses to a line segment on $y=2x$, with zero area. For $s<4$, orientation is preserved; for $s>4$, it is reversed. For example, at $s=1$ the image has area $3/2$.

EXERCISE 10.9 EXAM QUESTION

1. Let $A\in M_n(\mathbb{R})$ satisfy $\operatorname{rank}(A^2)<\operatorname{rank}(A)$. Please prove that there exists $v\in\mathbb{R}^n$ such that $A^2v=0$ and $Av\neq 0$.

2. Let $A,B\in M_n(\mathbb{R})$. Please prove that

$$\operatorname{rank}(A)+\operatorname{rank}(B)\leq n+\operatorname{rank}(AB).$$

   Please deduce that if $\operatorname{rank}(A)+\operatorname{rank}(B)>n$, then $AB\neq 0$.

3. Let $N\in M_n(\mathbb{R})$ satisfy $N^2=0$ and $\operatorname{rank}(N)=r$. Please find $\det(I_n+tN)$ and $\operatorname{rank}(I_n+tN)$ for every $t\in\mathbb{R}$.

WORKED SOLUTION

1. If $Av=0$, then $A^2v=A(Av)=0$. Hence $\ker A\subseteq\ker A^2$. By the rank-nullity theorem and the given inequality,

$$\begin{aligned}\dim\ker A^2 &= n-\operatorname{rank}(A^2)\\ &> n-\operatorname{rank}(A)=\dim\ker A.\end{aligned}$$

   Thus the inclusion is strict: $\ker A\subsetneq\ker A^2$. Choose $v\in\ker A^2\setminus\ker A$. Membership in the first kernel gives $A^2v=0$, while exclusion from the second gives $Av\neq 0$. This vector remains nonzero after one application of A and becomes zero after two applications.

2. Consider the restricted map

$$T:\operatorname{Im}B\longrightarrow\mathbb{R}^n,\qquad T(u)=Au.$$

   Its image is exactly $\operatorname{Im}(AB)$: every $u\in\operatorname{Im}B$ has the form $u=Bv$, so $T(u)=ABv$; conversely, every vector ABv arises in this way. Its kernel is $\operatorname{Im}B\cap\ker A$. Applying rank-nullity to T gives

$$\operatorname{rank}(B)=\dim(\operatorname{Im}B\cap\ker A)+\operatorname{rank}(AB).$$

   However,

$$\dim(\operatorname{Im}B\cap\ker A)\leq\dim\ker A=n-\operatorname{rank}(A).$$

   Substitute and rearrange:

$$\begin{aligned}\operatorname{rank}(B)&\leq n-\operatorname{rank}(A)+\operatorname{rank}(AB),\\ \operatorname{rank}(A)+\operatorname{rank}(B)&\leq n+\operatorname{rank}(AB).\end{aligned}$$

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EXERCISE 10.9 EXAM QUESTION CONTINUED

If the ranks sum to more than n, then $\operatorname{rank}(AB) > 0$, so AB is not the zero matrix.

3. First compute the two products:

$$(I_n + tN)(I_n - tN) = I_n - tN + tN - t^2N^2 = I_n,$$
$$(I_n - tN)(I_n + tN) = I_n + tN - tN - t^2N^2 = I_n.$$

Thus $I_n + tN$ is invertible for every $t \in \mathbb{R}$, with inverse $I_n - tN$. Immediately,

$$\operatorname{rank}(I_n + tN) = n.$$

To compute the determinant, note that $N^2 = 0$ implies $\operatorname{Im} N \subseteq \ker N$. Hence $r \leq n - r$, or $2r \leq n$. Choose a basis $(u_1, \ldots, u_r)$ of $\operatorname{Im} N$ and extend it to a basis $(u_1, \ldots, u_r, w_1, \ldots, w_{n-2r})$ of $\ker N$. For each j, choose v_j with $Nv_j = u_j$.

The list

$$\mathcal{B} = (u_1, \ldots, u_r, w_1, \ldots, w_{n-2r}, v_1, \ldots, v_r)$$

is a basis. Indeed, applying N to a vanishing linear combination gives a combination of the u_j, so every coefficient of a v_j is zero. The remaining coefficients vanish because they multiply a basis of $\ker N$. Thus the list contains n linearly independent vectors. In this basis,

$$[N]_{\mathcal{B}} = \begin{pmatrix} 0 & 0 & I_r \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad [I_n + tN]_{\mathcal{B}} = \begin{pmatrix} I_r & 0 & tI_r \\ 0 & I_{n-2r} & 0 \\ 0 & 0 & I_r \end{pmatrix}.$$

A block of size zero is simply omitted. This is upper triangular with every diagonal entry equal to 1. Since the determinant is preserved under change of basis,

$$\det(I_n + tN) = 1 \qquad \text{for every } t \in \mathbb{R}.$$

Neither answer depends on r: the determinant is 1 and the rank is n.

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Additional practice

EXERCISE 10.10 FOUNDATION

For $t \in \mathbb{R}$, consider

$$A_t = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 3 & 1 \\ 2 & 4 & t \end{pmatrix}.$$

1. Please compute $\det A_t$ using one row operation, and please determine when A_t is invertible.
2. At $t = 0$, please find a basis of $\ker A_0$ and please find $\operatorname{rank} A_0$.
3. Please construct B_t from A_t by, in this order, interchanging the first two rows, multiplying the third row by 2, and adding 5 times the third row to the first row. Please compute $\det B_t$ without writing B_t explicitly.

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EXERCISE 10.10 FOUNDATION CONTINUED

WORKED SOLUTION

1. The operation $R_3 \leftarrow R_3 - 2R_1$ preserves the determinant and gives

$$\begin{pmatrix} 1 & 2 & 0 \\ 0 & 3 & 1 \\ 0 & 0 & t \end{pmatrix}.$$

   Hence $\det A_t = 1 \cdot 3 \cdot t = 3t$, and A_t is invertible exactly when $t \neq 0$.

2. At $t = 0$, the kernel equations are $x_1 + 2x_2 = 0$ and $3x_2 + x_3 = 0$. Setting $x_2 = s$ gives

$$\ker A_0 = \{s(-2, 1, -3)^T : s \in \mathbb{R}\} = \operatorname{span}\{(-2, 1, -3)^T\}.$$

   The kernel has dimension 1, so rank-nullity gives $\operatorname{rank} A_0 = 3 - 1 = 2$.

3. The interchange multiplies the determinant by -1, scaling the row multiplies it by 2, and the final operation preserves it. Thus

$$\det B_t = (-1) \cdot 2 \cdot \det A_t = -6t.$$

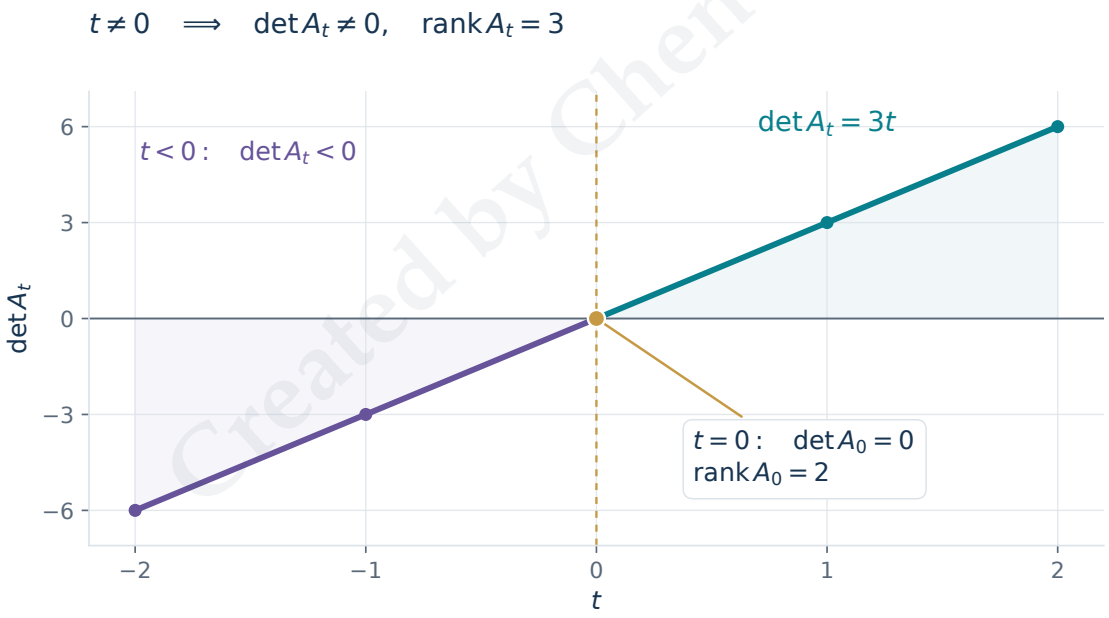


Figure 10.2. For the matrix A_t in the exercise, the graph $\det A_t = 3t$ crosses zero only at $t = 0$. For every $t \neq 0$, the matrix is invertible and has rank 3; at the gold point, $\operatorname{rank} A_0 = 2$, with kernel $\operatorname{span}\{(-2, 1, -3)^T\}$.

EXERCISE 10.11 CHALLENGE

Let $A \in M_n(\mathbb{F})$ be invertible, $U \in \mathbb{F}^{n \times k}$, and $V \in \mathbb{F}^{k \times n}$, with $n, k \geq 1$.

1. Please use determinants of block matrices to prove

$$\det(A + UV) = \det A \, \det(I_k + VA^{-1}U).$$

   Do not assume that U or V is square or invertible.

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EXERCISE 10.11 CHALLENGE CONTINUED

2. For $s \in \mathbb{R}$, please use the identity to compute the determinant of

$$C_s = \begin{pmatrix} s+1 & 0 & 1 \\ 0 & s+1 & -1 \\ s & s & 1 \end{pmatrix}$$

and please determine when it is invertible. At the singular parameter value, please find its kernel and rank.

WORKED SOLUTION

1. First prove $\det(I_n + XY) = \det(I_k + YX)$ for $X \in \mathbb{F}^{n\times k}$ and $Y \in \mathbb{F}^{k\times n}$. Set

$$K = \begin{pmatrix} I_n & X \\ -Y & I_k \end{pmatrix}, \quad L = \begin{pmatrix} I_n & 0 \\ Y & I_k \end{pmatrix}, \quad H = \begin{pmatrix} I_n & -X \\ 0 & I_k \end{pmatrix}.$$

Both L and H are block triangular with determinant 1. We have

$$LK = \begin{pmatrix} I_n & X \\ 0 & I_k + YX \end{pmatrix}, \qquad HK = \begin{pmatrix} I_n + XY & 0 \\ -Y & I_k \end{pmatrix}.$$

Multiplicativity and the determinant formula for block triangular matrices give

$$\det K = \det(I_k + YX) = \det(I_n + XY).$$

Now $A + UV = A(I_n + A^{-1}UV)$. Substitute $X = A^{-1}U$ and $Y = V$ to obtain the required identity.

2. Take $A = I_3$ and write $C_s = I_3 + UV$, where

$$U = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}, \qquad V = \begin{pmatrix} s & 0 & 1 \\ 0 & s & -1 \end{pmatrix}.$$

Direct multiplication verifies this factorization and gives

$$I_2 + VU = \begin{pmatrix} s+2 & 1 \\ -1 & s \end{pmatrix}.$$

Therefore

$$\det C_s = \det(I_2 + VU) = s(s+2) + 1 = (s+1)^2.$$

Thus C_s is invertible exactly when $s \neq -1$. At $s = -1$, its kernel equations are $x_3 = 0$ and $-x_1 - x_2 + x_3 = 0$, so

$$\ker C_{-1} = \operatorname{span}\{(1, -1, 0)^T\}, \qquad \operatorname{rank} C_{-1} = 3 - 1 = 2.$$

The multiplicity of a root of the determinant as a polynomial in s is not the kernel dimension: this root is double, while the kernel has dimension 1.

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EXERCISE 10.12 CHALLENGE

Let $A, B \in M_n(\mathbb{R})$ be invertible and satisfy $AB = -BA$. Please prove that n is even. For every $m \geq 1$, please construct such a pair of size $2m$ with $A^2 = B^2 = I_{2m}$.

WORKED SOLUTION

Determinant rules give

$$\det A \det B = \det(AB) = \det(-BA) = (-1)^n \det B \det A.$$

Both determinants are nonzero, so cancellation gives $1 = (-1)^n$. Over $\mathbb{R}$, n must be even. For every $m \geq 1$, take

$$A = \begin{pmatrix} I_m & 0 \\ 0 & -I_m \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & I_m \\ I_m & 0 \end{pmatrix}.$$

Direct block multiplication gives

$$AB = \begin{pmatrix} 0 & I_m \\ -I_m & 0 \end{pmatrix}, \qquad BA = \begin{pmatrix} 0 & -I_m \\ I_m & 0 \end{pmatrix} = -AB,$$

and $A^2 = B^2 = I_{2m}$. Thus both are invertible.

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EXERCISE 10.13 CHALLENGE

Let $n \geq 1$, and let $x_1, \ldots, x_n$ be distinct elements of a field F, and let $y_1, \ldots, y_n$ be distinct. Assume $x_i + y_j \neq 0$ for every i, j, and define

$$C = \left(\frac{1}{x_i + y_j} \right)_{i,j=1}^{n}.$$

Please prove that every square submatrix of C is invertible. Please deduce that the real matrix $H_n = (1/(i+j-1))_{i,j=1}^{n}$ is invertible for every $n \geq 1$.

WORKED SOLUTION

First suppose $Cc = 0$, and define

$$q(t) = \sum_{j=1}^{n} c_j \prod_{k \neq j} (t + y_k).$$

This polynomial has degree at most $n-1$. Row i of the equation gives

$$q(x_i) = \left(\prod_{k=1}^{n} (x_i + y_k) \right) \sum_{j=1}^{n} \frac{c_j}{x_i + y_j} = 0.$$

It has n distinct roots and must therefore be the zero polynomial. At $t = -y_j$, every term except term j vanishes, so

$$0 = q(-y_j) = c_j \prod_{k \neq j} (y_k - y_j).$$

The product is nonzero, forcing $c_j = 0$ for every j. Thus C is invertible. A square submatrix selects equally many row and column indices. The selected elements remain distinct within each list, and their sums are still nonzero, so the same proof applies. For the final assertion, take $x_i = i - 1$ and $y_j = j$ over $\mathbb{R}$. Every denominator $i + j - 1$ is positive, and $C = H_n$.

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